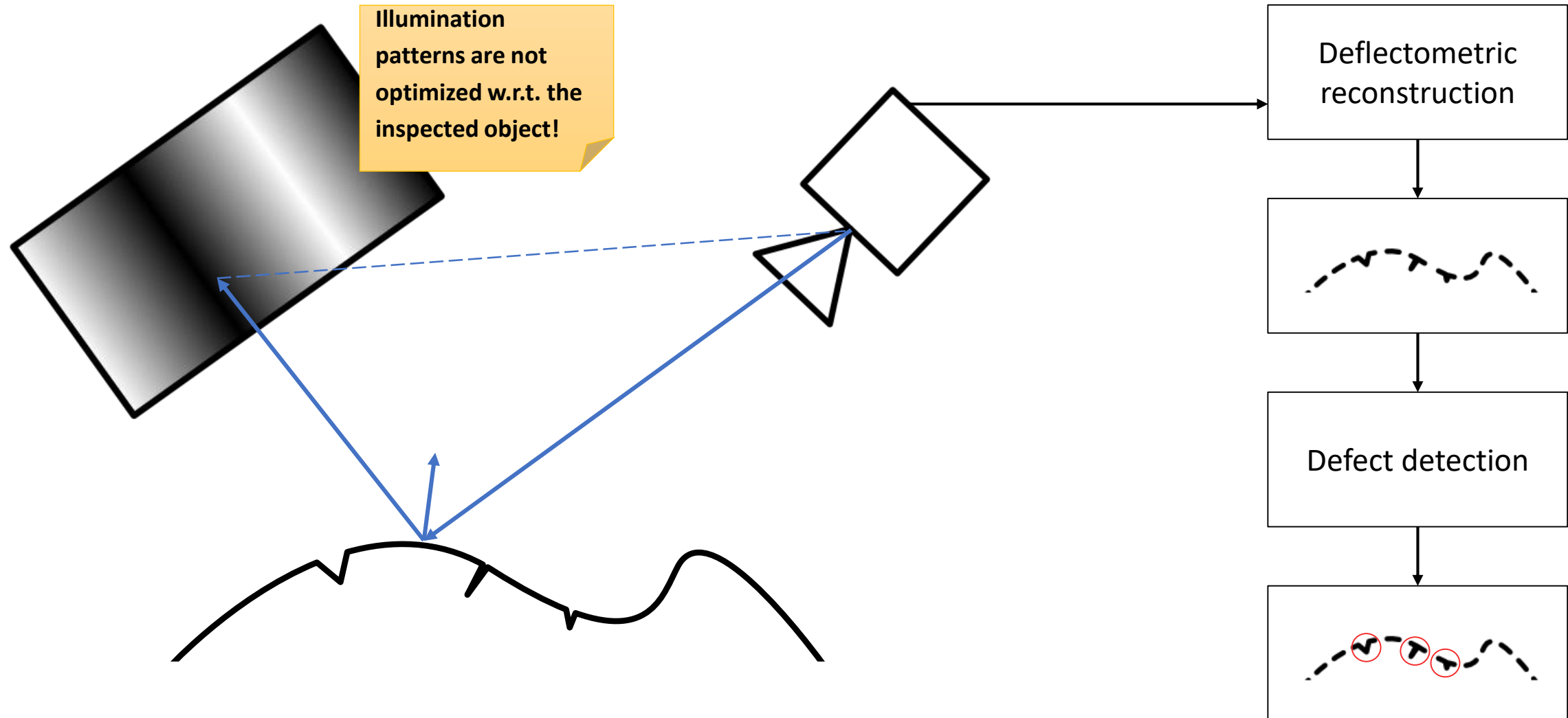


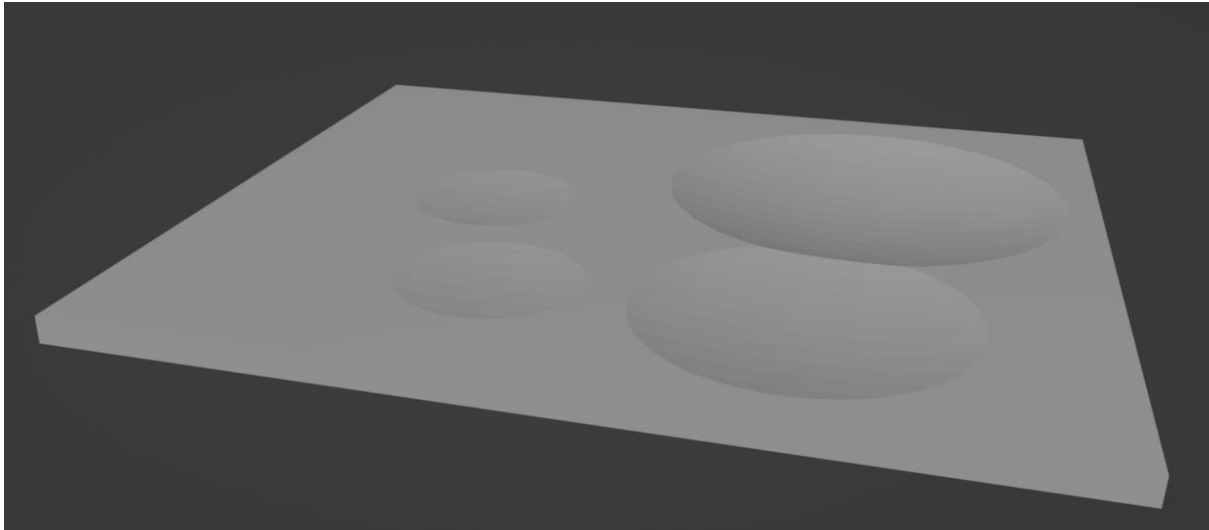
Inverse rendering for optimal illumination
patterns for deflectometric surface inspection

- Classic deflectometry: Fringes pattern → Decoding → Reconstruction → Defect detection



- Idea: Learn illumination pattern that is optimal for given object.
 - Instead of surface reconstruction
 - perform defect detection directly in camera image.

Test object: Flat mirror with parabolic valleys.

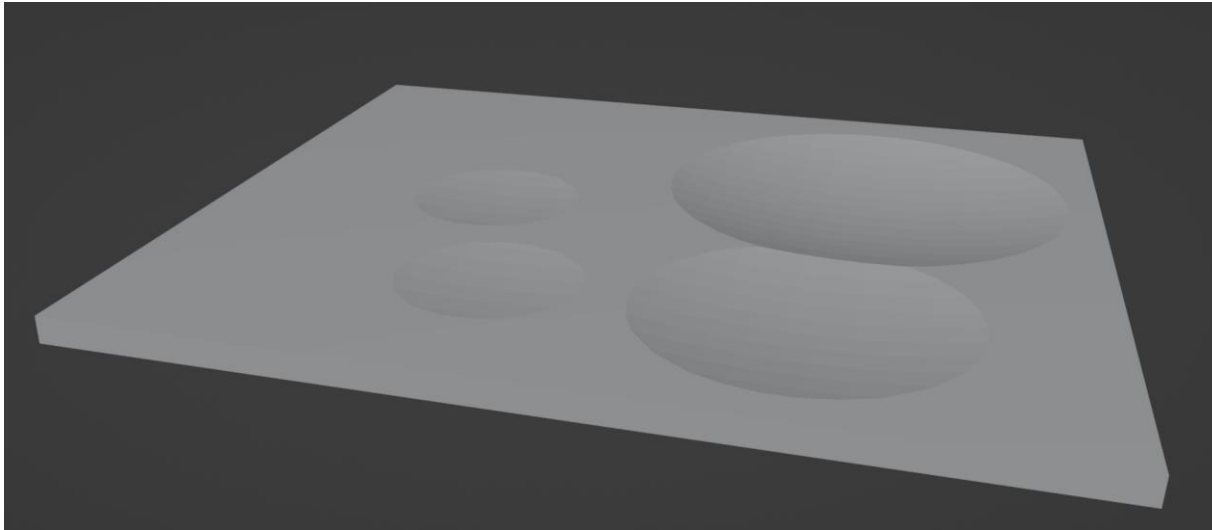


Generation of synthetic defects by boolean addition of spheres with mirroring surfaces



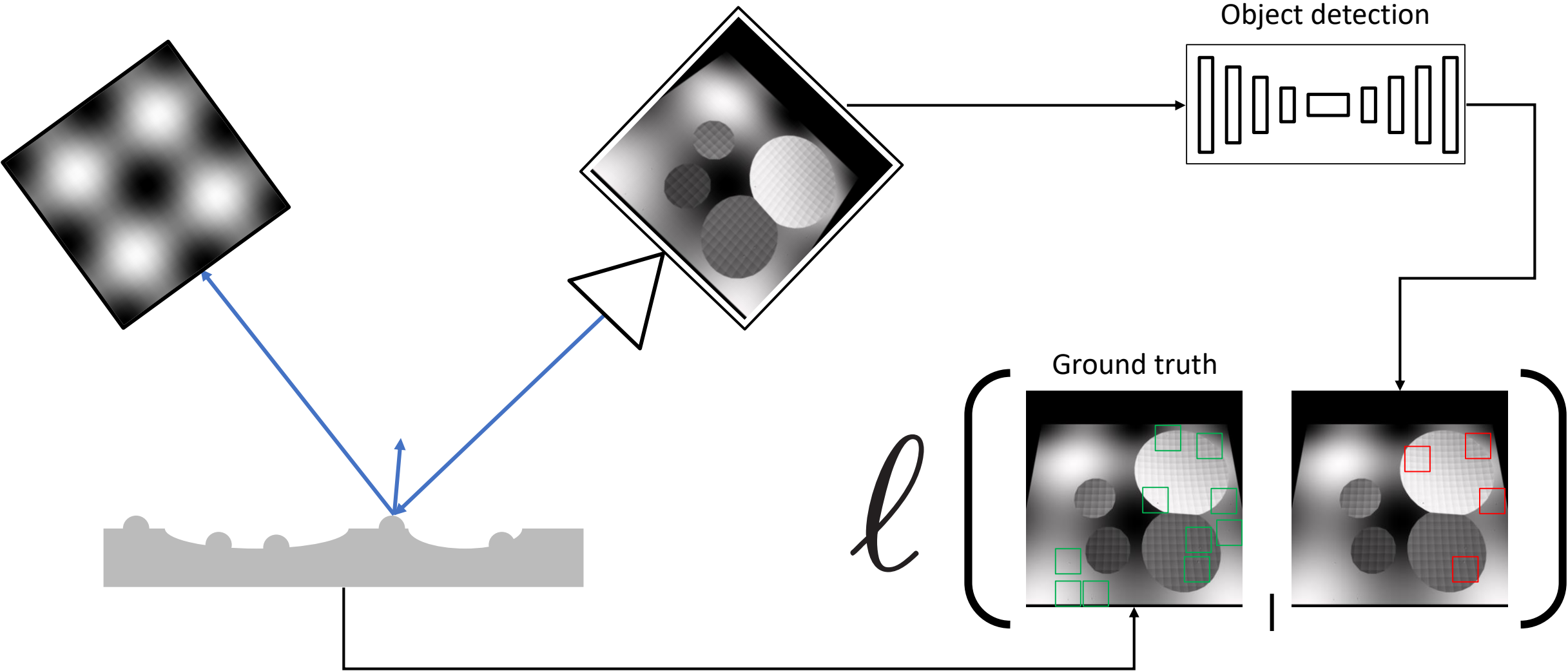
- Idea: Learn illumination pattern that is optimal for given object.
 - Instead of surface reconstruction
 - perform defect detection directly in camera image.

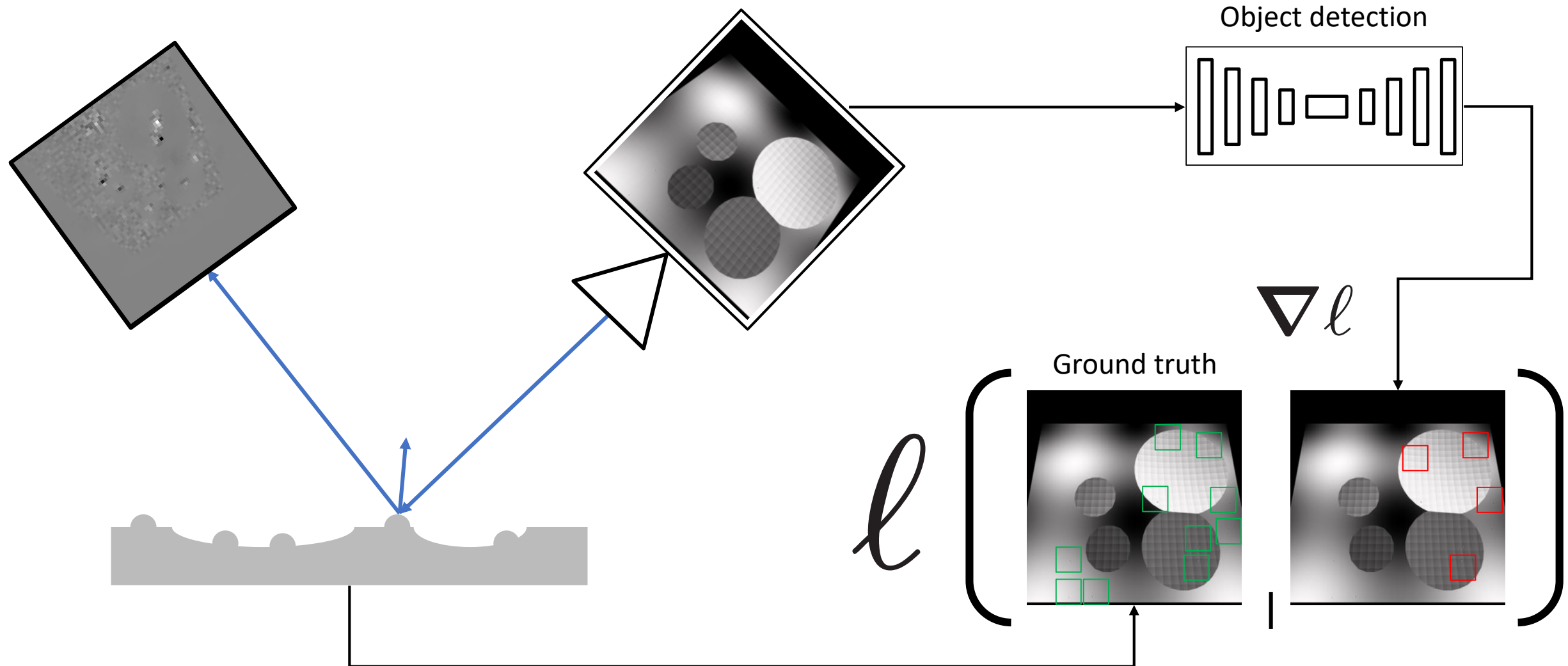
Test object: Flat mirror with parabolic valleys.



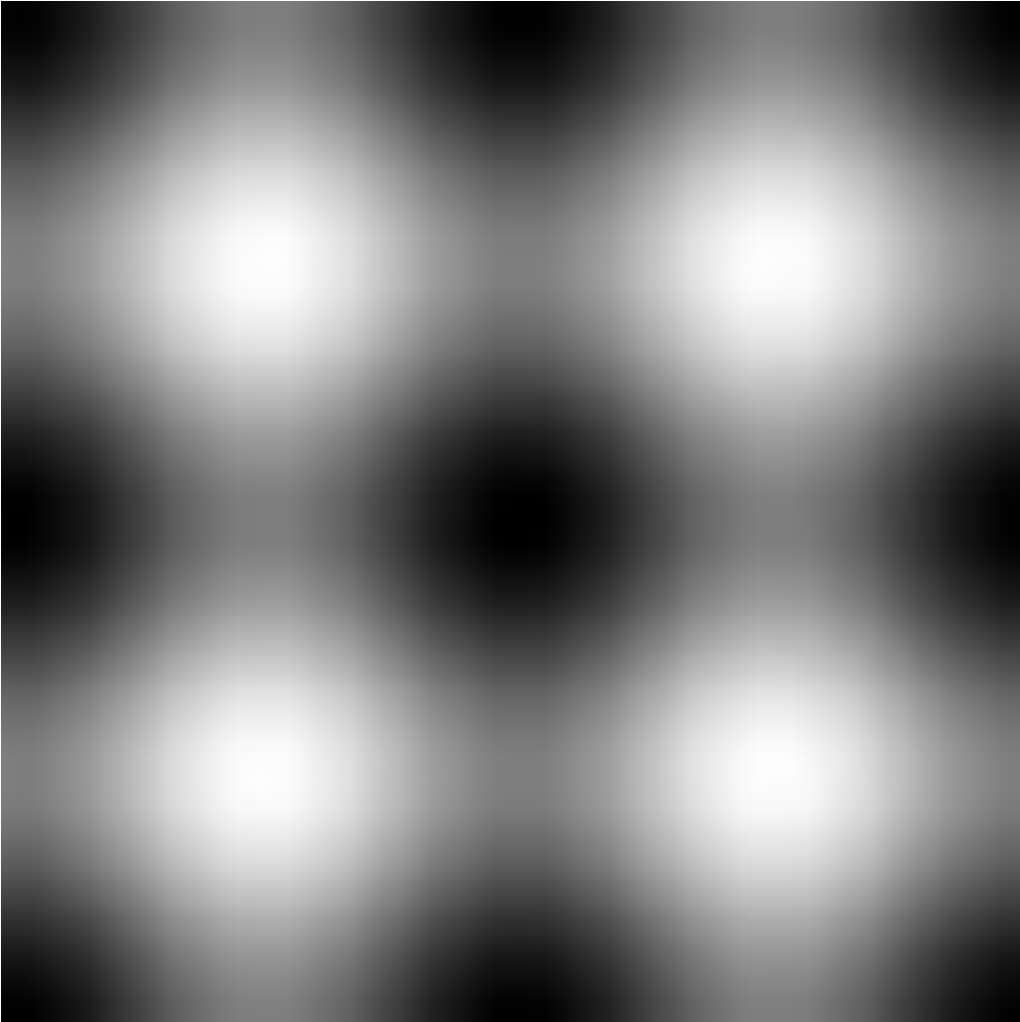
Generation of synthetic defects by boolean addition of spheres with mirroring surfaces







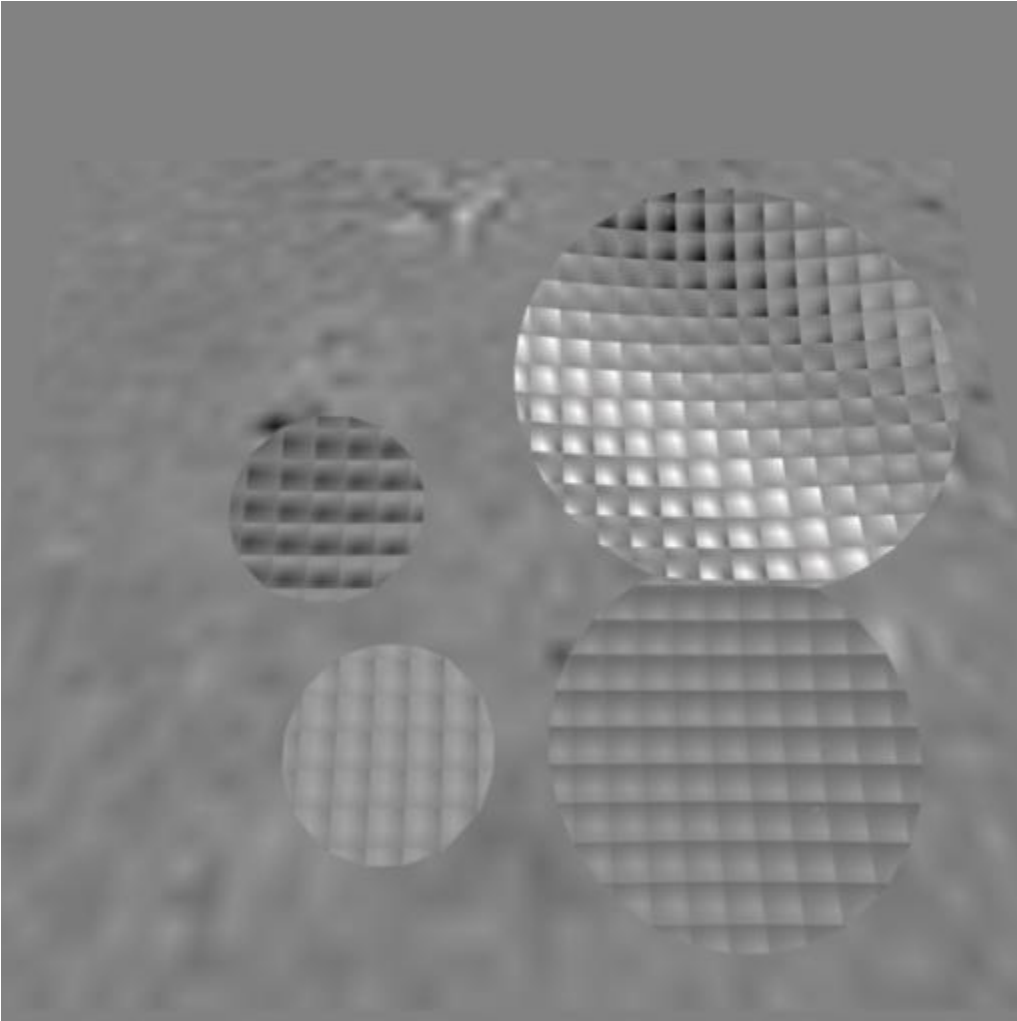
Initial pattern



Optimized pattern



Camera image



Optimized pattern

